S Moments A level only

Name:

Class:

Date:

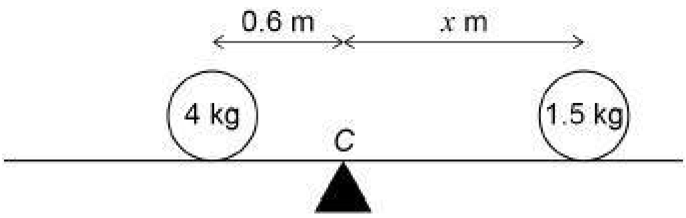
Time: **119 minutes**

Marks: **100 marks**

Comments:

Q1.

A uniform rod, *AB*, has length 4 metres.
The rod is resting on a support at its midpoint *C*.
A particle of mass 4 kg is placed 0.6 metres to the left of *C*.
Another particle of mass 1.5 kg is placed x metres to the right of *C*, as shown.



The rod is balanced in equilibrium at *C*.

Find x.

Circle your answer.

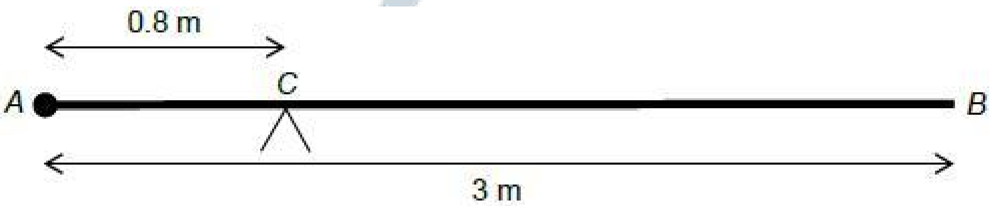
1.8 m   1.5 m   1.75 m   1.6 m

(Total 1 mark)

Q2.

A uniform rod, *AB*, has length 3 metres and mass 24 kg.

A particle of mass M kg is attached to the rod at *A*.
The rod is balanced in equilibrium on a support at *C*, which is 0.8 metres from *A*.

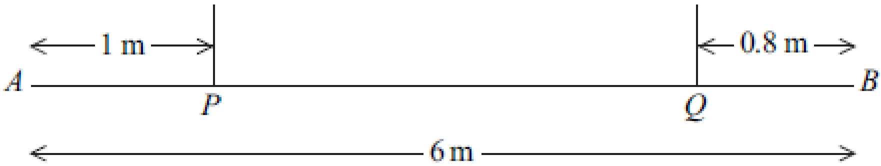


Find the value of M.

(Total 2 marks)

Q3.

A uniform plank AB, of length 6 m, has mass 25 kg. It is supported in equilibrium in a horizontal position by two vertical inextensible ropes. One of the ropes is attached to the plank at the point P and the other rope is attached to the plank at the point Q, where AP = 1 m and QB = 0.8 m, as shown in the diagram.



(a) (i) Find the tension in each rope.

(5)

(ii) State how you have used the fact that the plank is uniform in your solution.

(1)

(b) A particle of mass m kg is attached to the plank at point B, and the tension in each rope is now the same.

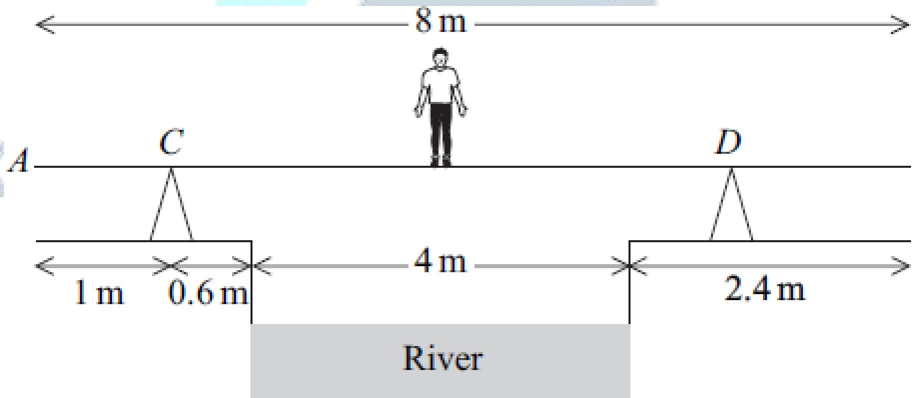
Find m.

(6)

(Total 12 marks)

Q4.

Ken is trying to cross a river of width 4 m. He has a uniform plank, AB, of length 8 m and mass 17 kg. The ground on both edges of the river bank is horizontal. The plank rests at two points, C and D, on fixed supports which are on opposite sides of the river. The plank is at right angles to both river banks and is horizontal. The distance AC is 1 m, and the point C is at a horizontal distance of 0.6 m from the river bank. Ken, who has mass 65 kg, stands on the plank directly above the middle of the river, as shown in the diagram.



(a) Draw a diagram to show the forces acting on the plank.

(2)

(b) Given that the reaction on the plank at the point D is $44g$ N, find the horizontal distance of the point D from the nearest river bank.

(4)

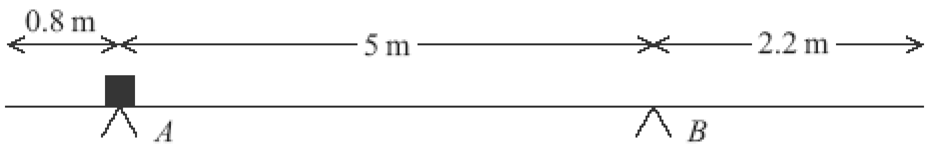
(c) State how you have used the fact that the plank is uniform in your solution.

(1)

(Total 7 marks)

Q5.

A uniform plank, of length 8 metres, has mass 30 kg. The plank is supported in equilibrium in a horizontal position by two smooth supports at the points A and B, as shown in the diagram. A block, of mass 20 kg, is placed on the plank at point A.



(a) Draw a diagram to show the forces acting on the plank.

(2)

(b) Show that the magnitude of the force exerted on the plank by the support at B is $19.2g$ newtons.

(3)

(c) Find the magnitude of the force exerted on the plank by the support at A.

(2)

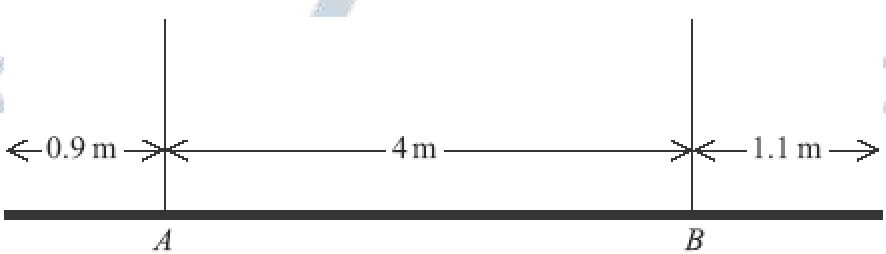
(d) Explain how you have used the fact that the plank is uniform in your solution.

(1)

(Total 8 marks)

Q6.

A uniform plank, of length 6 metres, has mass 40 kg. The plank is held in equilibrium in a horizontal position by two vertical ropes attached to the plank at A and B, as shown in the diagram.



(a) Draw a diagram to show the forces acting on the plank.

(1)

(b) Show that the tension in the rope attached to the plank at B is $21g$ N.

(3)

(c) Find the tension in the rope that is attached to the plank at A.

(2)

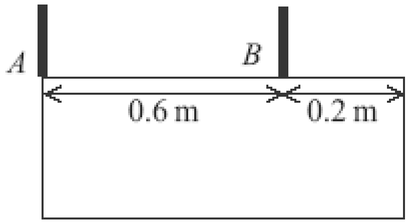
(d) State where in your solution you have used the fact that the plank is uniform.

(1)

(Total 7 marks)

Q7.

A hotel sign consists of a uniform rectangular lamina of weight W. The sign is suspended in equilibrium in a vertical plane by two vertical light chains attached to the sign at the points A and B, as shown in the diagram. The edge containing A and B is horizontal.



The tensions in the chains attached at A and B are T_A and T_B respectively.

(a) Draw a diagram to show the forces acting on the sign.

(1)

(b) Find T_A and T_B in terms of W.

(4)

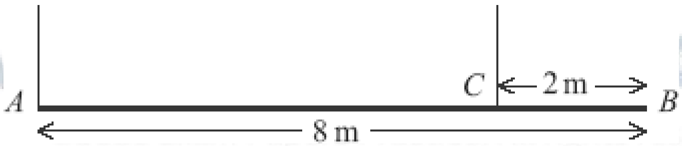
(c) Explain how you have used the fact that the lamina is uniform in answering part (b).

(1)

(Total 6 marks)

Q8.

A uniform plank, AB, is 8 m long and has mass 30 kg. It is supported in equilibrium in a horizontal position by two vertical inextensible ropes. One of the ropes is attached to the plank at A and the other rope to the point C, where BC = 2 m, as shown in the diagram.

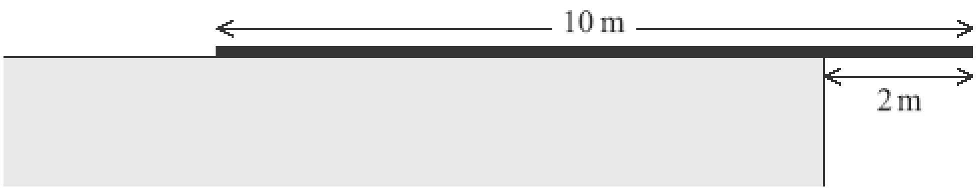


Find the tension in each rope.

(Total 5 marks)

Q9.

A uniform plank is 10 m long and has mass 15 kg. It is placed on horizontal ground at the edge of a vertical river bank, so that 2 m of the plank is projecting over the edge, as shown in the diagram below.



(a) A woman of mass 50 kg stands on the part of the plank which projects over the river.

Find the greatest distance from the river bank at which she can safely stand.

(3)

(b) The woman wishes to stand safely at the end of the plank which projects over the river.

Find the minimum mass which she should place on the other end of the plank so that she can do this.

(4)

(c) State how you have used the fact that the plank is uniform in your solution.

(1)

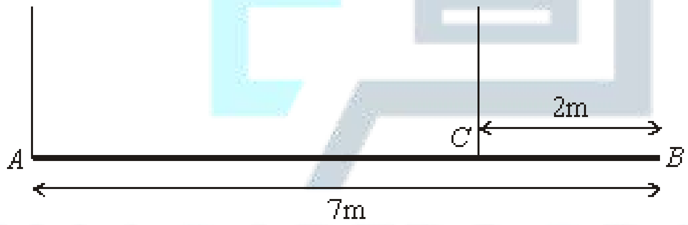
(d) State one other modelling assumption which you have made.

(1)

(Total 9 marks)

Q10.

A uniform beam, AB, has mass 20 kg and length 7 metres. A rope is attached to the beam at A. A second rope is attached to the beam at the point C, which is 2 metres from B. Both of the ropes are vertical. The beam is in equilibrium in a horizontal position, as shown in the diagram.

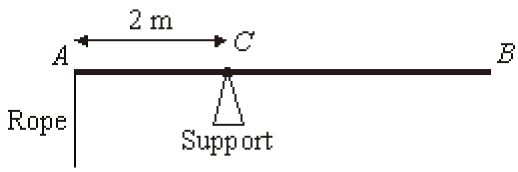


Find the tensions in the two ropes.

(Total 6 marks)

Q11.

The diagram shows a uniform rod, AB, of mass 10 kg and length 5 metres. The rod is held in equilibrium in a horizontal position, by a support at C and a light vertical rope attached to A, where AC is 2 metres.



(a) Draw and label a diagram to show the forces acting on the rod.

(1)

(b) Show that the tension in the rope is 24.5 N.

(3)

(c) A package of mass m kg is suspended from B. The tension in the rope has to be doubled to maintain equilibrium.

(i) Find m.

(4)

(ii) Find the magnitude of the force exerted on the rod by the support.

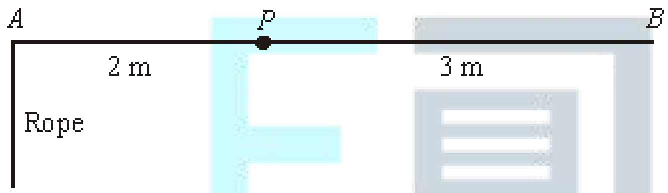
(3)

(d) Explain how you have used the fact that the rod is uniform in your solution.

(1)
(Total 12 marks)

Q12.

A uniform beam AB has length 5 metres and mass 10 kg. It is freely pivoted at the point P which is 2 metres from . A rope is attached to the beam at A and exerts a vertical force on the beam at A. The beam and rope are shown in the diagram.



(a) The beam is in equilibrium and at rest in a horizontal position.

(i) Show that the tension in the rope is 24.5 N.

(3)

(ii) A particle of mass 40 kg is then fixed to the beam at B. The tension in the rope is increased, so that the beam remains horizontal. Find the tension in the rope in this case.

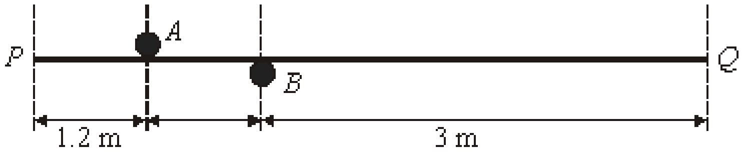
(3)

(b) How would your answer to part (a)(ii) change if the beam had been in equilibrium at an angle to the horizontal, but with the rope still vertical? Explain why.

(2)
(Total 8 marks)

Q13.

A uniform beam, PQ, has length 5 metres and mass 40 kg. It is placed between two fixed horizontal bars, A and B, so that the beam remains horizontal, as shown in the diagram.



(a) Draw a diagram to show the forces acting on the beam.

(1)

(b) Show that the force exerted on the beam by bar A has magnitude 245 N.

(3)

(c) Find the magnitude of the force exerted on the beam by bar B.

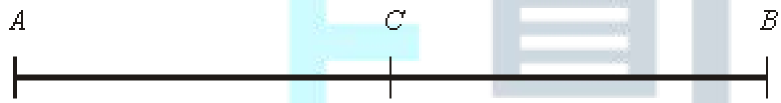
(2)

(d) An object of mass 5 kg is placed on the beam at Q. Find the forces now exerted on the beam by the bars A and B.

(6)
(Total 12 marks)

Q14.

A uniform rod, AB, has length 4 metres, and C is the mid-point of AB.



Three particles are attached to the rod:

one of mass 7 kg at A;
one of mass 12 kg at C;
and one of mass 11 kg at B.

The mass of the uniform rod AB is 10 kg.

The rod rests in a horizontal position supported at A and B.

Find the magnitudes of the vertical forces exerted by the supports on the rod at A and B.

(Total 5 marks)